Toward a Decisive Test of Energy Conservation in Driven Nonequilibrium Electromagnetic Systems: Residual Work, Entropy Accounting, and EVRT-Class Anomaly Criteria

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Abstract

This work establishes a stringent experimental and thermodynamic framework for evaluating whether driven nonequilibrium electromagnetic systems can produce persistent residual energy or mechanical output beyond conventional energy accounting. Building on Emergent Vacuum Response Theory (EVRT), a complete energy-budget formalism is combined with entropy-consistent analysis, uncertainty propagation, and artifact-controlled measurement protocols. The objective is not to assert violations of conservation laws, but to define the precise experimental conditions under which such a claim would become scientifically meaningful. A set of falsifiable criteria is developed for identifying residual energy-transfer effects that cannot be explained within standard thermodynamic and electromagnetic frameworks.

1 Introduction

High-voltage, asymmetric electromagnetic systems have historically produced small force signals that are often attributed to known physical mechanisms such as electrostatic coupling, ion wind, thermal drift, and electromagnetic interactions. Distinguishing genuine physical effects from artifacts requires complete energy accounting and rigorous experimental validation.

This work defines a decisive test framework specifying the conditions under which residual energy or force measurements could challenge conventional interpretations of energy conservation.

2 Energy Budget Framework

The governing equation for system-level energy accounting is:

$$P_{\text{res}} = P_{\text{in}} - \left(\frac{dU_{\text{stored}}}{dt} + P_{\text{mech}} + P_{\text{loss}} \right) \tag{1}$$

A null-consistent system satisfies:

$$P_{\text{res}} = 0 \pm \sigma_{\text{res}} \tag{2}$$

3 Residual Significance Criterion

A residual is considered statistically meaningful only if:

$$|P_{\text{res}}| > n\sigma_{\text{res}} \tag{3}$$

where $n \geq 3$ for preliminary significance.

4 Thermodynamic Constraints

The system must remain consistent with:

$$\Delta G = \Delta H - T\Delta S \tag{4}$$

and

$$\frac{dS_{\text{total}}}{dt} \geq 0 \tag{5}$$

These conditions enforce that any observed effect cannot violate fundamental thermodynamic principles without requiring a revision of physical theory.

5 Entropy Closure Requirement

A complete energy-budget analysis must include entropy accounting. Any apparent residual must be evaluated against possible entropy-producing processes, including:

- Joule heating
- dielectric relaxation
- radiation losses
- environmental coupling

Failure to close entropy accounting invalidates any claim of anomalous energy transfer.

6 Calorimetric Validation

Independent calorimetric measurement provides a critical cross-check:

$$Q_{\text{measured}} \approx \int P_{\text{loss}} dt \tag{6}$$

Agreement between electrical input, thermal output, and mechanical work is required to validate energy closure.

7 Criteria for a Physics-Challenging Result

A result may be considered physics-challenging only if all of the following are satisfied:

- $|P_{\text{res}}| > n\sigma_{\text{res}}$
- Reproducibility across independent runs
- Persistence under null configurations
- Survival of polarity reversal and symmetry tests
- No correlation with environmental variables
- Independent replication by separate apparatus
- Verified calorimetric consistency
- No identifiable entropy deficit

8 Failure Modes

Common artifact sources include:

- Ion wind and corona discharge
- Thermal expansion and convection
- Electrostatic coupling to surroundings
- RF leakage and induced currents
- Mechanical vibration and acoustic coupling

Each must be explicitly tested and eliminated.

9 Apparatus Schematic

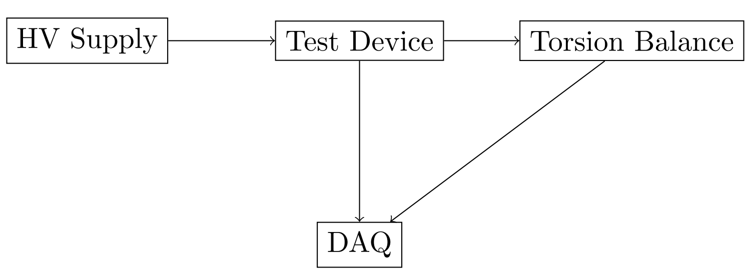

Figure 1: Simplified apparatus schematic.

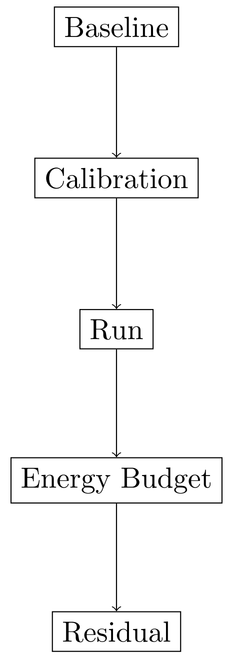

Figure 2: Measurement and analysis workflow.

10 Measurement Workflow

11 Interpretation

A positive residual does not imply free energy. Instead, it indicates:

- incomplete system modeling
- unaccounted energy pathways
- or previously uncharacterized coupling mechanisms

12 Discussion

This framework reframes anomalous-force experiments as precision energy-accounting problems. By enforcing thermodynamic consistency and uncertainty-bounded measurement, the protocol ensures that any claimed anomaly must withstand rigorous scrutiny.

13 Conclusion

This work defines the experimental and thermodynamic conditions required to evaluate extreme claims regarding energy transfer in nonequilibrium electromagnetic systems. The framework provides a conservative and falsifiable pathway for identifying residual effects while maintaining consistency with established physical laws.